IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

SOFTWARE SYSTEMS

Friday, 10 May 2:30 pm

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Baig
Second Marker(s) : J. Pitt

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Software Systems

Marks: 60
Time: 2 hrs

Question 1

(a) Draw an Entity Relationship diagram for the following description of a bookstore database. Show the entities, relationships, cardinalities of relationships, participation types of relationships, attributes, and primary keys.

Each book in the system contains a title, a unique ISBN, a price, and number of copies sold. Each book also has a discounted price which is half the original price if the number of copies sold is less than ten, and equal to the original price otherwise. Each author contains a name, not necessarily unique, and one or more addresses. Each publisher contains a unique name, one or more addresses, and one or more telephone numbers. Each customer contains a name and a unique email address. Books may have individual authors or several co-authors. Each author and publisher could have several books in the system. However, there may be authors and publishers in the system whose books currently do not exist in the system. Every book has exactly one publisher. Authors of books and their publishers together promote the books. A book may be promoted any number of times and authors and publishers may participate in any number of promotions. Customers purchase books at specific dates. A customer may recommend other customers, but a customer may not be recommended more than once. From time to time, a customer may also receive vouchers. No more than one voucher is given at a time. A voucher is for a certain amount and has a start and end date.

[10]

(b) Convert the Entity Relationship model from (a) into a relational schema. Highlight the primary key of each relation in the schema.

[5]

(c) Consider the following relational schema of actors, plays, and the characters played by actors in the plays.

Actor (<u>actorID</u>, name, dateOfBirth)
Play (<u>playID</u>, title, author, yearWritten)
Character (<u>actorID</u>, <u>playID</u>, <u>characterName</u>)

Write SQL queries to perform the following tasks:

i. Find the names of actors who have acted in at least one play written by ‘Chekov’ and never acted in a play written by ‘Arthur Miller’.
ii. Find IDs of actors who have played characters in three or more different plays written by ‘Shakespeare’. (Note: an actor might have played more than one character in the same play, in which case that play should only be counted once.)

[5 + 5]

Question 2

(a) Mark the following statements as true or false. Explain (in a line or two) the reasons behind your answers:

i. Transmission delay is independent of the length of the physical link.
ii. Queueing delays (due to queueing of data packets) are not a concern in circuit switched networks.
iii. For a fixed network path with varying load the processing delay per packet is generally constant over time however the propagation delay per packet may vary.

[2 + 2 + 2]

(b) Suppose a browser is using HTTP to fetch a web page with a base html file of 30, 000 bits and the following six objects embedded in the page:

ob1.jpg	15,000 bits
ob2.jpg	5000 bits
ob3.jpg	10,000 bits
ob4.jpg	10,000 bits
ob5.txt	5,000 bits
ob6.jpg	35,000 bits

All senders use a transport layer window size of 20, 000 bits. For this problem, ignore the overhead of header related bytes. Consider all transmitted data to be application data.

i. How many RTTs would it take for the transfer of the base html file to the browser?

How many RTTs would it take to download the complete page, including all embedded objects, when using:

ii. non-persistent HTPP?
iii. non-persistent HTPP with 4 parallel connections? (Assume that the 4 parallel connections can be established in parallel).
iv. persistent HTPP with pipelining but no parallelism?

[2 + 2 + 2 + 2]

(c) Consider the Go-Back-N protocol with a sender window size of N. Suppose that at time t, the next in-order packet that the receiver is expecting has a sequence number of k. Assume that the medium does not reorder packets and the sequence number space is sufficiently large. Give the set of all possible sequence numbers that could be part of the sender's window at time t. Justify your answer.

[3]

(d) Consider transferring a 4 GB file from host X to host Y. The maximum segment size of the transport layer protocol is 512 bytes. Before a packet is sent out from host X, a total of 64 bytes of transport, network and data link layer headers are also added to the packet. The packet is then sent out over a 256 Mbps link. Assuming that X can send packets back-to-back and continuously, how long will it take for the entire file to be transmitted?

[3]

Question 3

(a) Suppose you connect to a network and DHCP provides you with the following information:

- IP address: 192.168.0.4
- Network Mask: 255.255.255.128
- Gateway: 192.168.0.1

i. If your routing table has only one entry, what is the IP address of the next hop in this entry?

ii. Of all possible IP addresses your computer may want to talk to, for which range will your computer not use this entry in the routing table?

Besides your own computer, there are the following three nodes connected to the network:

node	IP	MAC
router	192.168.0.1	F3-00-21-00-13-44
node1	192.168.0.35	EB-01-06-00-23-12
node2	192.168.0.154	EB-01-06-00-24-AB

Suppose your ARP table is initially empty, and you send two IP packets, one for node 1 and another one for node 2.

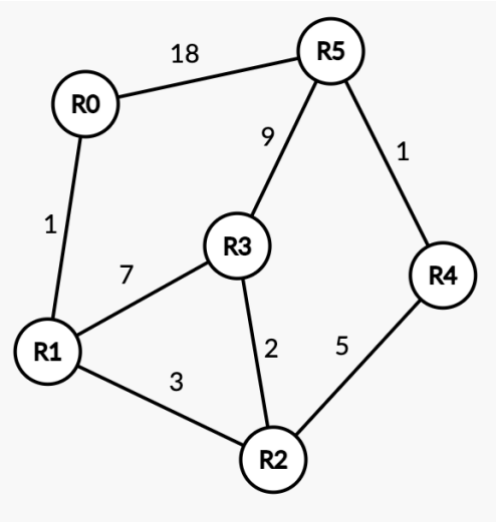
iii. What are the IP routes (sequence of IP addresses visited by the packet, including the source and destination) taken by each of these two packets?

(Assume that the ARP protocol was used as required to make the communication possible.)

iv. What are the two entries in the ARP table after these two packets are sent?

[1 + 2 + 2 + 2]

(b) Consider the following network of routers R0, R1, …, R5 and the links between them with their associated costs:



Suppose the network uses a Distance Vector routing algorithm. For this problem, we focus only on paths to destination R5 from sources R0, …, R4. In the following table, there is a column for each source router. Each entry in these columns shows the distance to R5 on the best path computed so far from that source, and the next hop router on that path. We ignore the parts of distance vectors for destinations other than R5.

Each row corresponds to an iteration of the algorithm. The rightmost column shows which nodes have transmitted their distance vectors to their neighbours in this iteration. On iteration # 1 R5 transmits its vector to its neighbours. Generally, every node whose vector gets updated transmits its vector to its neighbours in the next iteration.

The first two rows of the table have been completed for you. Add rows to this table until further iterations do not update any vector. Also write the transmitters in each iteration.

Iteration	R0	R1	R2	R3	R4	Transmitters
0	∞	∞	∞	∞	∞	-
1	18, R5	∞	∞	9, R5	1, R5	R5

[4]

(c) Differentiate between the following terms:

i. LANs and Autonomous Systems
ii. Iterative and recursive DNS queries

[2 + 2]

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